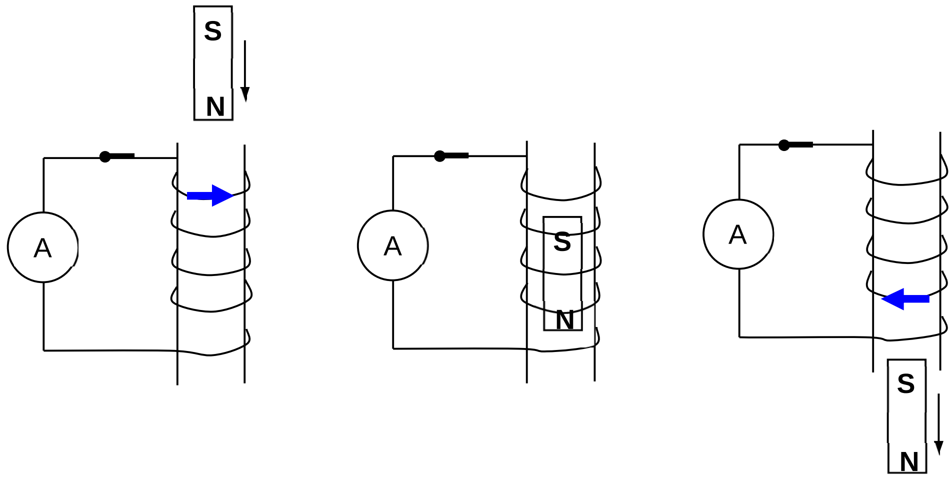
$Q = 135$ J

6 **(a)** State Lenz's Law.

Lenz's law states that the e.m.f. is induced in the direction such that its effects oppose the change in the magnetic flux linkage that is inducing it. [1]

A strong magnet is allowed to drop through a solenoid which is wound around a cardboard tube. The solenoid is connected to a switch and an ammeter.



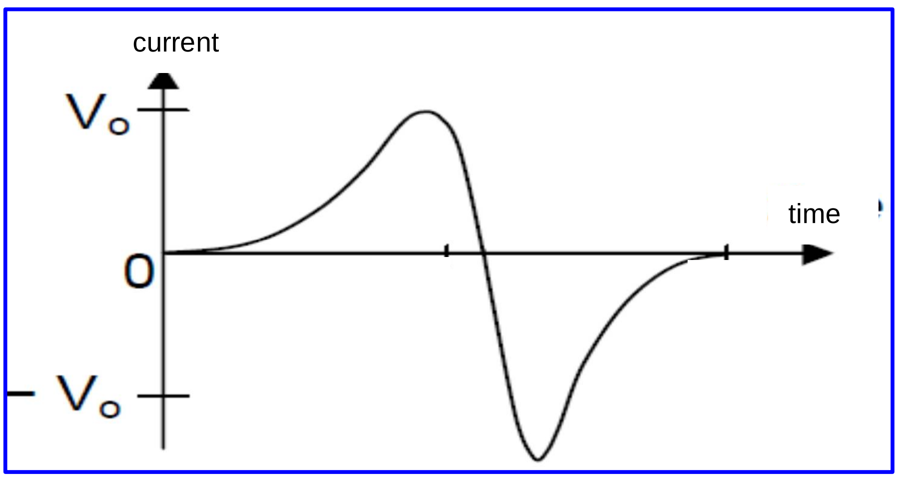
Magnet approaching solenoid Fig. 6.1A

Magnet in the middle of solenoid Fig. 6.1B

Magnet leaving solenoid Fig. 6.1C

(b) Indicate, with arrows on the winding of the solenoid in Fig. 6.1A, Fig. 6.1B and Fig. 6.1C, the direction of any induced current. [2]

(c) Hence, sketch on Fig. 6.2 a labelled graph showing the variation with time of the current induced in the solenoid as the magnet falls through it. [2]



Correct shape/positive & negative current. [1]

Peak of current for retreating magnet is greater. [1]

(d) The switch connected to the solenoid is now opened, and the same magnet is allowed to drop through the solenoid again.

State and explain any change in the speed of the magnet as it leaves the solenoid.

The speed of the magnet increases. [1]

Since the switch is opened; there is no induced current in the solenoid to produce an opposing induced magnetic field to oppose the motion of the magnet. [1]

7 **(a)** Tungsten, a transition metal, is commonly used as a target metal for the production of